Business Forecasting – PDF Code Collection

Code snippets extracted from TYDS Business Forecasting Practical.pdf. Explanations and output sections are omitted.

1. Program:

```
expert_opinions <- c(500, 550, 520, 530, 540)

# Average forecast
forecast <- mean(expert_opinions)

print(paste("Forecasted Demand:", forecast))
```

2. R Program

```
# Historical sales data
sales <- c(120, 135, 150, 145, 160, 170)

# 3-month moving average forecast
forecast <- mean(tail(sales, 3))

print(paste("Forecast for Next Month:", forecast, "units\n"))
```

3. Program:

```
student_id <- c(101,102,103,104,105,106,107,108,109,110)

student_name <- c("Amit","Priya","Rahul","Sneha",
          "Vikas","Neha","Ajay","Pooja",
          "Rohan","Anjali")

marks <- c(85,72,91,65,78,88,55,92,70,81)

student_data <- data.frame(student_id,
                student_name,
                marks)

student_data$grade <- ifelse(student_data$marks >= 80, "A",
             ifelse(student_data$marks >= 60,
                  "B","C"))

print(student_data)

summary(student_data$marks)

grade_count <- table(student_data$grade)

print(grade_count)

barplot(grade_count,
     main="Grade Distribution",
     col=c("green","yellow","red"))

pie(grade_count,
   main="Grade Distribution",
   col=c("green","yellow","red"))

hist(student_data$marks,
   main="Marks Distribution",
   col="skyblue")

boxplot(student_data$marks,
    main="Marks Boxplot",
    col="orange")
```

4. R Code

```
# Student Marks Data Collection

student_id <- c(101,102,103,104,105,106,107,108,109,110)

student_name <- c("Amit","Priya","Rahul","Sneha",
          "Vikas","Neha","Ajay","Pooja",
          "Rohan","Anjali")

marks <- c(85,72,91,65,78,88,55,92,70,81)

student_data <- data.frame(student_id,
                student_name,
                marks)

print(student_data)
```

5. R Code

```
student_data$grade <- ifelse(student_data$marks >= 80, "A",
            ifelse(student_data$marks >= 60,
                 "B","C"))

print(student_data)
```

6. R Code

```
grade_count <- table(student_data$grade)

barplot(grade_count,
    main="Grade Distribution",
    xlab="Grade",
    ylab="Number of Students",
    col=c("green","yellow","red"))
```

7. R Code

```
pie(grade_count,
  main="Student Grade Distribution",
  col=c("green","yellow","red"))
```

8. R Code

```
hist(student_data$marks,
   main="Distribution of Marks",
   xlab="Marks",
   col="skyblue",
   border="black")
```

9. R Code

```
boxplot(student_data$marks,
    main="Student Marks Boxplot",
    col="orange")
```

10. Program:

```
# Monthly Sales Data (24 Months)



sales <- c(

    100,120,140,180,200,220,240,230,210,260,280,320,

    110,130,150,190,210,240,260,250,230,280,300,340

)



# Create Time Series Object

sales_ts <- ts(sales,

           start = c(2024,1),

           frequency = 12)



# Display Time Series

print(sales_ts)



# Plot Original Series

plot(sales_ts,

     main = "Monthly Sales Time Series",

     xlab = "Month",

     ylab = "Sales",

     col = "blue",

     lwd = 2)
# Decompose Time Series
decomp <- decompose(sales_ts)
```

```
# Plot Components
plot(decomp)

# Extract Components
trend <- decomp$trend
seasonal <- decomp$seasonal
random <- decomp$random

# Print Components
cat("Trend Component:\n")
print(trend)

cat("\nSeasonal Component:\n")
print(seasonal)

cat("\nRandom Component:\n")
print(random)

# Moving Average Trend
trend_ma <- filter(sales_ts,
                rep(1/3,3),
                sides = 2)

# Plot Trend
plot(sales_ts,
   main = "Trend Analysis using Moving Average",
   xlab = "Month",
   ylab = "Sales")


    col = "red",
    lwd = 3)

legend("topleft",
    legend = c("Original Data","Trend"),
    col = c("black","red"),
    lty = 1,
    lwd = 2)
```

Year	Sales
2021	1000
2022	1200
2023	1500

2024 1800

Month	Sales
Jan	100
Feb	120
Mar	140
Apr	180
May	200
Jun	220
Jul	240
Aug	230
Sep	210
Oct	260
Nov	280
Dec	320

```
sales <- c(100,120,140,180,200,220,
```

```
      240,230,210,260,280,320)


sales_ts <- ts(sales,

         start=c(2025,1),

         frequency=12)


print(sales_ts)



plot(sales_ts,

  main="Monthly Mobile Sales",

  xlab="Month",

  ylab="Sales (Thousands)",

  col="blue",

  lwd=2)
```

11. Code

```
# ----------------------------------------
# Practical: Linear Trend Analysis in R
# Business Scenario:
# Predict food deliveries for July
# ----------------------------------------

# Step 1: Create the dataset
Month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun")
Deliveries <- c(8200, 8500, 8900, 9300, 9700, 10100)

# Step 2: Assign X values (Time)
X <- 1:6

# Step 3: Create Data Frame
delivery_data <- data.frame(Month, X, Deliveries)

# Display Dataset
print("Food Delivery Dataset")
print(delivery_data)

# Step 4: Apply Linear Trend Method
trend_model <- lm(Deliveries ~ X)

# Step 5: Display Trend Equation
print("Linear Trend Model")
print(summary(trend_model))

# Extract coefficients
a <- coef(trend_model)[1]
b <- coef(trend_model)[2]

cat("\nTrend Equation:\n")
cat("Y =", round(a,2), "+", round(b,2), "* X\n")

# Step 6: Predict Deliveries for July (X = 7)
july_prediction <- predict(trend_model,
                  newdata = data.frame(X = 7))

cat("\nPredicted Deliveries for July =",
  round(july_prediction), "\n")

# Step 7: Plot Actual Data and Trend Line
plot(X, Deliveries,
   pch = 19,
   col = "blue",
   xlab = "Month Number",
   ylab = "Daily Deliveries",
   main = "Linear Trend Analysis")

abline(trend_model,
     col = "red",
     lwd = 2)

# Add July Prediction
points(7, july_prediction,
    pch = 19,
```

```
    col = "green",
    cex = 1.5)

text(7, july_prediction,
   labels = "July Forecast",
   pos = 3,
   col = "green")

# Step 8: Business Decision
  cat("\nBusiness Recommendation:\n")
  cat("The number of deliveries is expected to increase.\n")
  cat("The company should hire additional delivery partners.\n")
  cat("\nBusiness Recommendation:\n")
  cat("Current number of delivery partners is sufficient.\n")
}
```

12. R Program

```
#----------------------------------------
# Seasonal Variation - Simple Average Method
#----------------------------------------

# Step 1: Enter quarterly sales data
sales <- c(120,180,160,140,
       130,195,170,150)

# Display original data
print("Quarterly Sales Data")
print(sales)

# Step 2: Calculate Overall Average
overall_average <- mean(sales)

cat("\nOverall Average =", overall_average,"\n")

# Step 3: Arrange data into matrix
sales_matrix <- matrix(sales,
              nrow = 2,
              byrow = TRUE)

colnames(sales_matrix) <- c("Q1","Q2","Q3","Q4")
rownames(sales_matrix) <- c("Year1","Year2")

print("\nSales Matrix")
print(sales_matrix)

# Step 4: Calculate Seasonal Average
seasonal_average <- colMeans(sales_matrix)

cat("\nSeasonal Average\n")
print(seasonal_average)

# Step 5: Calculate Seasonal Index
seasonal_index <- (seasonal_average / overall_average) * 100

cat("\nSeasonal Index\n")
print(round(seasonal_index,2))

# Step 6: Interpretation
cat("\nInterpretation:\n")

  if(seasonal_index[i] > 100)
  {
    cat(colnames(sales_matrix)[i],
       "- Sales are ABOVE average.\n")
  }
  else
  {
    cat(colnames(sales_matrix)[i],
       "- Sales are BELOW average.\n")
  }
}
```

13. Program:

```
# ------------------------------------------
# LINEAR REGRESSION - E-COMMERCE CASE STUDY
# ------------------------------------------

# Step 1: Create the dataset

advertising <- c(10, 12, 15, 18, 20, 22, 25, 28, 30, 32)

sales <- c(50, 55, 60, 65, 70, 75, 80, 85, 90, 95)

# Create data frame
ecommerce <- data.frame(advertising, sales)
```

```
# Display data
print(ecommerce)


# Step 2: Create Linear Regression Model
# sales is dependent variable
# advertising is independent variable

model <- lm(sales ~ advertising, data = ecommerce)


# Step 3: Display regression model

print(model)


# Step 4: Display detailed summary

summary(model)


# Step 5: Get regression coefficients

coefficients(model)


# Step 6: Predict sales when advertising
# expenditure is ■35,000

new_data <- data.frame(advertising = 35)

predicted_sales <- predict(model, newdata = new_data)

print(predicted_sales)


# Step 7: Display prediction in a meaningful format

cat("Advertising Expenditure: ■35,000\n")
cat("Predicted Sales Revenue: ■",
  round(predicted_sales, 2), " thousand\n")


# Step 8: Plot the original data

plot(
  ecommerce$advertising,
  ecommerce$sales,
  main = "Advertising Expenditure vs Sales",
  xlab = "Advertising Expenditure (■000)",
  ylab = "Sales Revenue (■000)",
  pch = 19
)


# Step 9: Add regression line

abline(model, lwd = 2)


# Step 10: Predict sales for multiple advertising budgets

future_advertising <- data.frame(
  advertising = c(15, 20, 25, 30, 35, 40)
)

future_predictions <- predict(
  model,
  newdata = future_advertising
)

prediction_table <- data.frame(
  Advertising = future_advertising$advertising,
  Predicted_Sales = round(future_predictions, 2)
)

print(prediction_table)
```

14. Program:

```
# Historical sales data

month <- c(1, 2, 3, 4, 5, 6)

sales <- c(120, 135, 150, 165, 180, 195)

# Create data frame
```

```
data <- data.frame(month, sales)

# Display data

print(data)

# Create linear regression model

model <- lm(sales ~ month, data = data)

# Display regression results

summary(model)

# Display coefficients

coefficients(model)

# Forecast July

july <- data.frame(month = 7)

forecast <- predict(model, newdata = july)

# Display forecast

cat("Forecasted Sales for July =",   forecast, "thousand\n")
```

15. Code

```
#
# SALES FORECASTING USING SEASONAL
# Ratio-to-Average Method
#

# Historical quarterly sales data
# Sales are in ■ lakh

Q1 <- c(80, 100)
Q2 <- c(120, 150)
Q3 <- c(160, 200)
Q4 <- c(100, 125)

# Create data frame
sales_data <- data.frame(
  Year = c("Year 1", "Year 2"),

    Q1 = Q1,
    Q2 = Q2,
    Q3 = Q3,
    Q4 = Q4
)

# Display historical data
print(sales_data)



#
# STEP 1: Calculate Annual Sales
#

sales_data$Annual_Sales <- rowSums(
  sales_data[, c("Q1", "Q2", "Q3", "Q4")]
)

print(sales_data)



#

# STEP 2: Calculate Annual Average
#

sales_data$Annual_Average <-

print(sales_data)



#
# STEP 3: Calculate Seasonal Relatives
# Formula:
# Seasonal Relative = Quarterly Sales /
#
seasonal_relative <- data.frame(
```

```
  Q1 = sales_data$Q1 /
  Q2 = sales_data$Q2 /

  Q3 = sales_data$Q3 /
  Q4 = sales_data$Q4 /
)

cat("\nSeasonal Relatives:\n")
print(seasonal_relative)



#
# STEP 4: Calculate Seasonal Indices
# Average seasonal relative for each
#

seasonal_index <-
colMeans(seasonal_relative)

cat("\nSeasonal Indices:\n")
print(seasonal_index)

#
# STEP 5: Calculate Year 3 Trend
#

year1_sales <- sales_data$Annual_Sales[1]
year2_sales <- sales_data$Annual_Sales[2]

# Increase in annual sales
annual_increase <- year2_sales -

# Forecast Year 3 annual sales
year3_sales <- year2_sales +

cat("\nYear 1 Sales =", year1_sales,
cat("Year 2 Sales =", year2_sales,
cat("Annual Increase =", annual_increase,
cat("Forecasted Year 3 Sales =",

#
# STEP 6: Calculate Average Quarterly
#

year3_quarterly_average <- year3_sales /

cat(
  "Year 3 Average Quarterly Sales =",
  year3_quarterly_average,
  "lakh\n"
)



#
# STEP 7: Forecast Year 3 Quarterly Sales
#
# Forecast = Trend Average × Seasonal

#

forecast_Q1 <- year3_quarterly_average *
  seasonal_index["Q1"] / 100

forecast_Q2 <- year3_quarterly_average *
  seasonal_index["Q2"] / 100

forecast_Q3 <- year3_quarterly_average *
  seasonal_index["Q3"] / 100

forecast_Q4 <- year3_quarterly_average *
  seasonal_index["Q4"] / 100



#
# STEP 8: Display Forecast
#

forecast <- data.frame(
  Quarter = c("Q1", "Q2", "Q3", "Q4"),

  Seasonal_Index =
  Forecast_Sales = c(
    forecast_Q1,
    forecast_Q2,
    forecast_Q3,
```

```
    forecast_Q4
  )
)

cat("\nYear 3 Sales Forecast:\n")
print(forecast)



#
# STEP 9: Calculate Total Forecast
#

total_forecast <-
sum(forecast$Forecast_Sales)

cat(
  "\nTotal Forecasted Sales for Year 3

    total_forecast,
    "lakh\n"
)
```